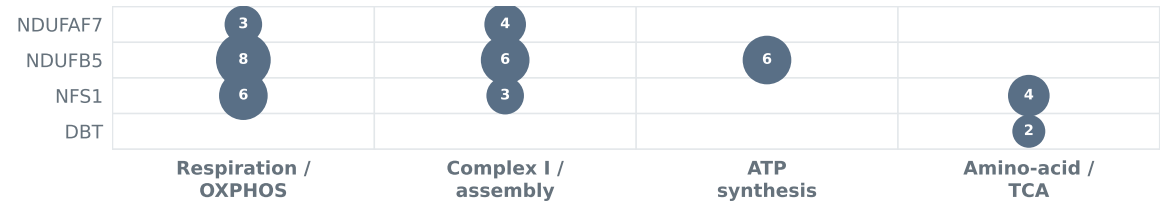
Figure 7. Biology of cross-omic discoveries

Mitochondrial leading edges, AREA-only neuronal signaling, and recurrent gene functions

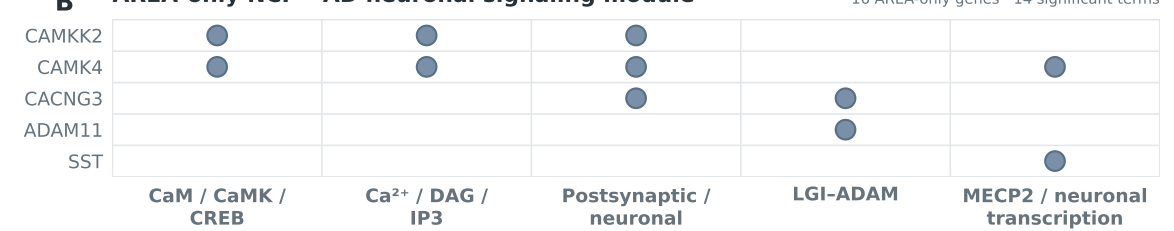
A

Shared mitochondrial leading-edge genes



B

AREA-only NCI → AD neuronal signaling module



C

Functional annotations of recurrent genes

Gene	Curated function / molecular activity
KIF13B	Involved in reorganization of the cortical cytoskeleton.
CSRP1	actinin binding [GO:0042805]
VGF	Neurosecretory protein VGF: Secreted polyprotein that is packaged and proteolytically processed by prohormone convertases PCSK1 and PCSK2 in a cell-type-specific manner (By similarity).
SST	Somatostatin-14: Inhibits the secretion of pituitary hormones, including that of growth hormone/somatotropin (GH1), PRL, ACTH, luteinizing hormone (LH) and TSH.
NRIP2	Down-regulates transcriptional activation by nuclear receptors such as NR1F2 (By similarity).
RAB9A	RAB9A is involved in the transport of proteins between the endosomes and the trans-Golgi network (TGN).
PPEF1	Dephosphorylates and destabilises PDCD5 to inhibit TP53/p53-mediated signaling.
SLC38A2	Symporter that cotransports neutral amino acids and sodium ions from the extracellular to the intracellular side of the cell membrane.
DHRS7B	Reductase required for adipogenesis and activation of PPARG nuclear receptor (By similarity).
TAPT1	Plays a role in primary cilia formation.

Function excerpts: UniProtKB/Swiss-Prot (NCBI Gene fallback). GO molecular function is used for tentative summaries.
Source qualifiers are retained; full annotations, GO terms, accessions and retrieval dates are saved with the figure.

A: leading-edge membership across concordant RNA–protein pathways. B: focal genes in significant AREA-only enrichment terms.